

```
In [1]: import os
        print(os.listdir())
```

Out:

```
['.ipynb_checkpoints', 'IRIS.csv', 'Untitled.ipynb']
```

```
In [2]: import pandas as pd
        import numpy as np

        import matplotlib.pyplot as plt
        import seaborn as sns

        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import LabelEncoder
        from sklearn.ensemble import RandomForestClassifier

        from sklearn.metrics import (
            accuracy_score,
            classification_report,
            confusion_matrix
        )
```

```
In [3]: df = pd.read_csv("IRIS.csv")

        df.head()
```

Out:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

```
In [4]: print("Shape:", df.shape)
```

```
df.info()
```

Out:

```
Shape: (150, 5)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   sepal_length    150 non-null    float64
 1   sepal_width     150 non-null    float64
 2   petal_length    150 non-null    float64
 3   petal_width     150 non-null    float64
 4   species         150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
In [5]: print(df.columns.tolist())
```

Out:

```
['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'species']
```

```
In [6]: df.isnull().sum()
```

Out:

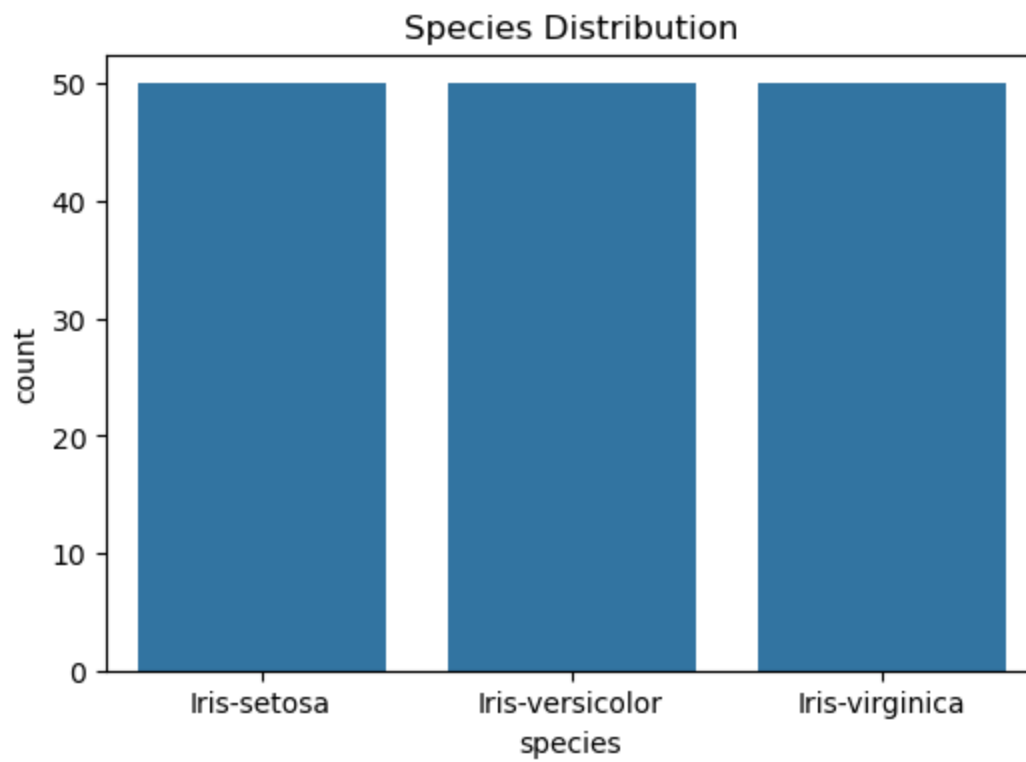
```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

```
In [7]: plt.figure(figsize=(6,4))
```

```
sns.countplot(x='species', data=df)
```

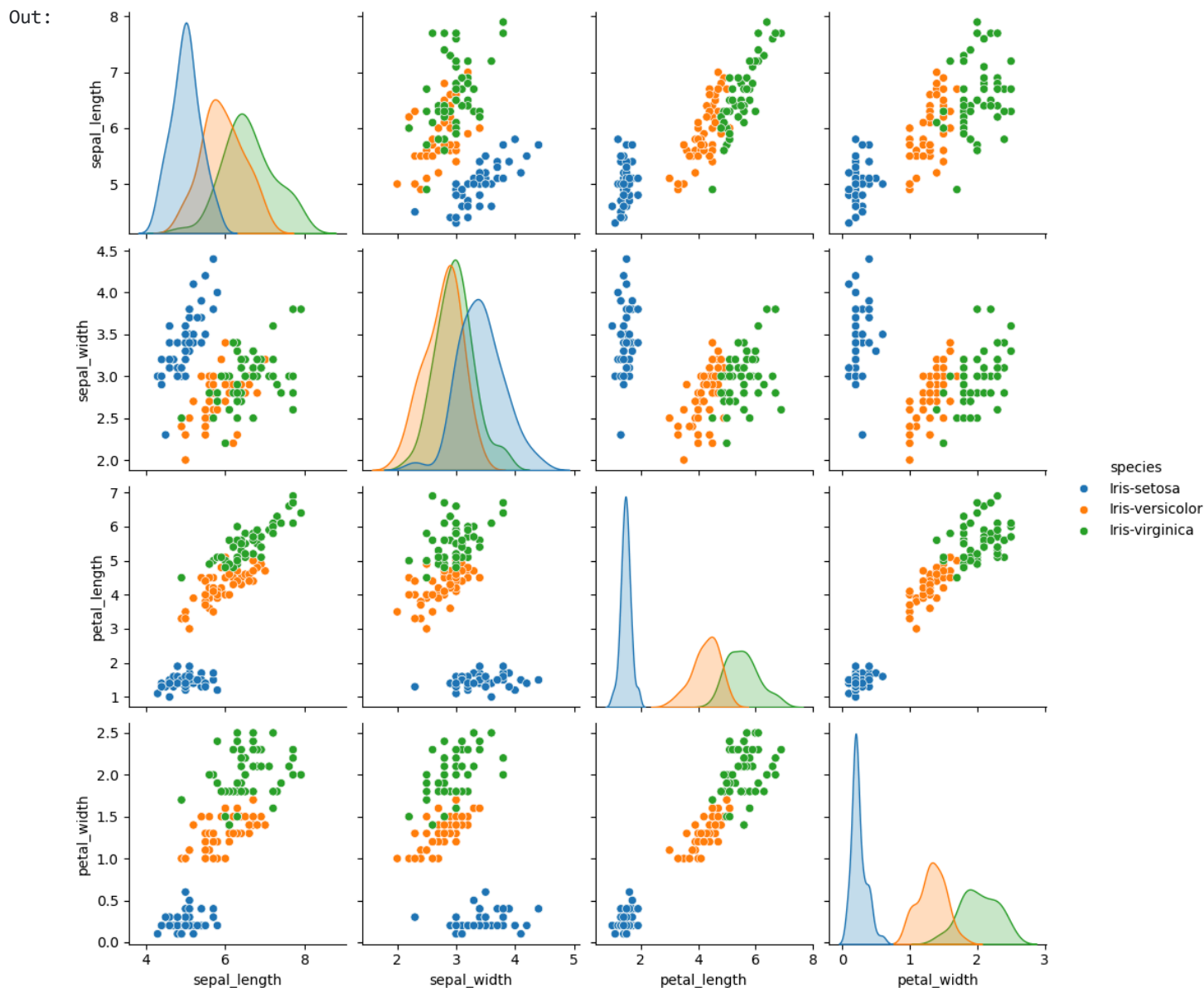
```
plt.title("Species Distribution")  
plt.show()
```

Out:



```
In [8]: sns.pairplot(df, hue='species')
```

```
plt.show()
```

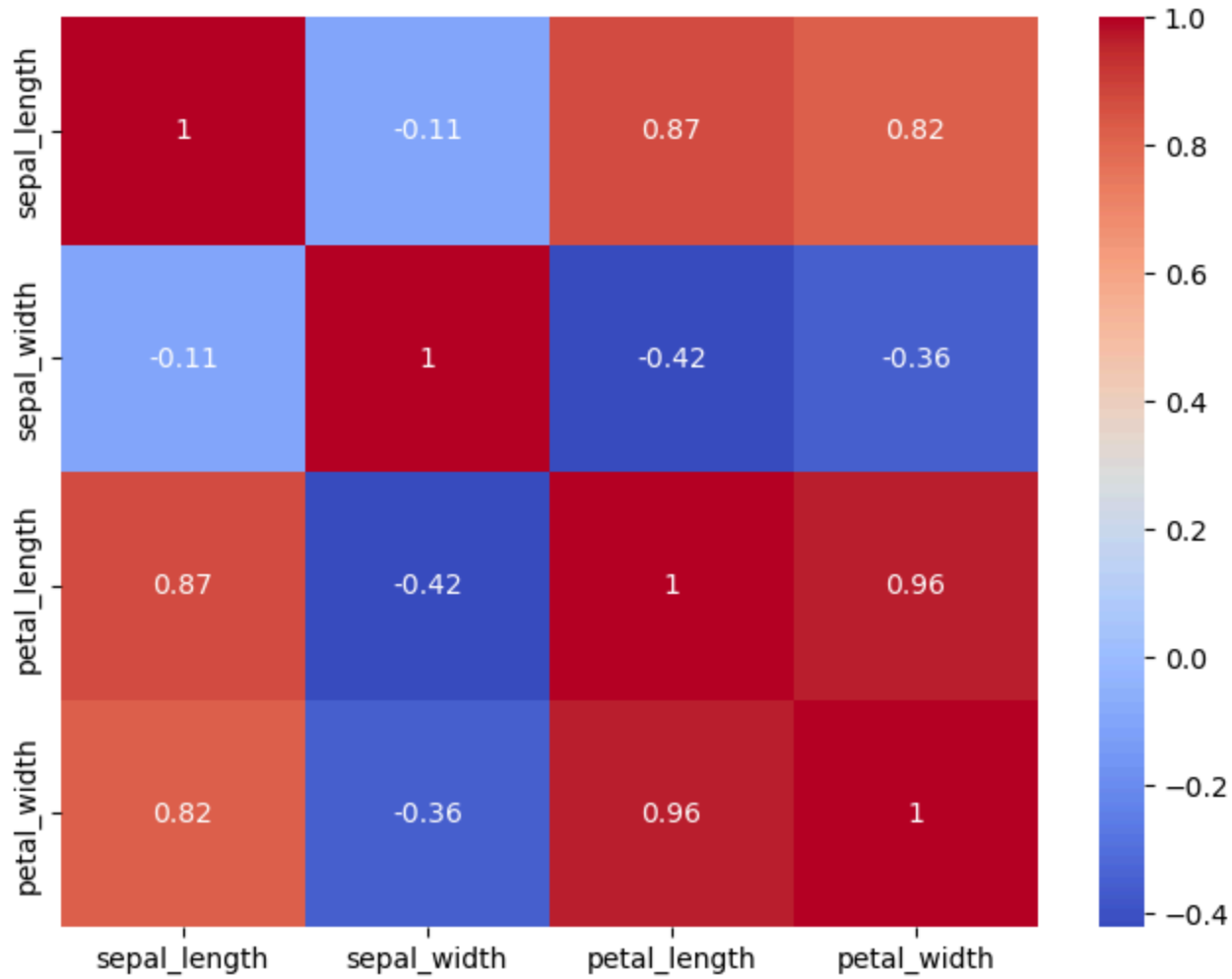


```
In [9]: plt.figure(figsize=(8,6))

sns.heatmap(
    df.select_dtypes(include=np.number).corr(),
    annot=True,
    cmap='coolwarm'
)

plt.show()
```

Out:



```
In [10]: le = LabelEncoder()

df['species'] = le.fit_transform(df['species'])

df.head()
```

Out:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

```
In [11]: X = df.drop('species', axis=1)
```

```
y = df['species']
```

```
In [12]: X_train, X_test, y_train, y_test = train_test_split(  
        X,  
        y,  
        test_size=0.2,  
        random_state=42  
    )
```